

EDGE MINDS HACKATHON 2026

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Project Title

Autonomous Local Research Agent Using a Small Language Model

Introduction

Research tasks often require more than a single prompt. A useful research system must break a topic into smaller questions, search for relevant information, filter useful evidence, summarize the findings, and compile them into a structured final report. The attached presentation defines this as an autonomous local research agent in which an SLM plans sub-questions, searches local text files, writes evidence-backed summaries, and produces a final report without intermediate user guidance. This approach is valuable because it keeps knowledge local, improves auditability, and supports privacy-sensitive use cases where external APIs may not be suitable.

Objectives

The main objective of this project is to design a lightweight AI research assistant in which the SLM acts as the central reasoning engine. The system should be able to generate sub-questions, search a local folder of text files, extract relevant passages, summarize findings for each sub-question, and compile a coherent final report. Another objective is to demonstrate true agentic behavior, where the model dynamically decides what to do next instead of following a fixed one-pass pipeline.

Methodology

The project follows a multi-step architecture shown in the presentation. First, the user provides a research topic. The SLM planner then creates 3 to 6 sub-questions and determines the order of investigation. A local folder search tool scans text files, retrieves relevant snippets, and returns file references. A summarization tool produces a focused answer for each question, and a report compiler merges all section summaries into the final report. The process repeats until the system determines that all important sub-questions have been covered.

Scope

This project is suitable for local document analysis, academic assistance, offline knowledge retrieval, and private research workflows. It can later be extended with chunking, embeddings, vector search, confidence scoring, and stronger state tracking for unanswered gaps.